

Tear Down a Board Game SOP (1x Episode: One Finished Game)

One episode tears down one finished board game and packs it back into its box. The sixteen pawns and the four score markers go into four color bags. The center tile, the twelve coins, the four dice and the first player marker go into the neutral bag. The twenty-four cards are gathered into one face down deck and held by one elastic band. The six empty trays are nested into one stack. The board is folded to four layers. Everything goes into the box base, the lid goes on, and the closed box is set flat on the output shelf. Then the episode ends.

The order never changes: color bags, neutral bag, cards, trays, board, pack the box, lid, shelf. The color order never changes either: red first, then blue, then green, then yellow. One bag is filled and sealed before the next one is started.

The left gripper works the red and blue player areas and the rules sheet. The right gripper works the green and yellow player areas, the bank tray and the dice well. Both grippers work together on the center tile, the elastic band, the finished tray stack, the folded board, the lid and the closed box. Only one loose piece is picked up at a time. No pawn, marker, coin or die is ever carried two at a time.

The table is set up in one of three ways. Only the output shelf moves, and the box base moves with it. The box base always stands on the table directly in front of the shelf, and the lid lies open side up beside the box base. The board area, the player row, the bag rack, the banding cradle, the nesting spot, the rules spot, the bank spot and the dice spot are in the same place in all three.

- **Config L:** the output shelf stands at the left end of the back edge. The box base stands on the table directly in front of it, and the lid lies to the right of the box base.
- **Config M:** the output shelf stands in the middle of the back edge. The box base stands on the table directly in front of it, and the lid lies to the left of the box base.
- **Config R:** the output shelf stands at the right end of the back edge. The box base stands on the table directly in front of it, and the lid lies to the left of the box base.

The steps are the same in every config, and each arm keeps the same role. Only the place the box is packed, the place the lid comes from and the place the closed box goes change, so only the direction of those carries changes. Steps 6, 7 and 8 say where to look.

What stays the same in every session:

- **Start position:** the output shelf stands at the left end of the back edge (**Config L**), in the middle of the back edge (**Config M**), or at the right end of the back edge (**Config R**). One config per episode, picked before recording and never changed during the episode. The rest of the back edge is clear.

- **Fixed order:** Steps 1 to 9 are done in number order. A step starts only after the step before it has reached its end state.
- **Fixed color order:** red, blue, green, yellow. Fill and seal one color bag before starting the next color.
- **What goes in a color bag:** exactly four pawns of that color and the one score marker of that color. Nothing else.
- **What goes in the neutral bag:** the center tile first, then the twelve coins, then the four dice, then the first player marker last. Eighteen pieces in all.
- **Card order:** gather the four card stacks from the red, blue, green and yellow trays in that order, then the eight card draw pile. The finished deck holds twenty-four cards, all face down.
- **Tray order:** from the bottom up, bank tray, red tray, blue tray, green tray, yellow tray, dice well.
- **Board folds:** the long fold first, the short fold second. The playing face ends up inside the folded board.
- **Packing order:** the folded board, the rules sheet, the tray stack, the banded deck, then the five sealed bags in the order neutral, red, blue, green, yellow.
- **One loose piece at a time:** a gripper never holds two pawns, two markers, two coins or two dice at once.
- **Lift clear, then move:** every piece is lifted clear of what it was sitting on before it travels, and piles, trays, the board, the lid and the closed box are kept level.
- **Lower, then let go:** a gripper lets go only after the thing it carries is resting and still.
- **Retry limit:** at most one deliberate re-grip and at most two small nudges for the same piece or stack.
- **Final state:** the lid is fully on, the closed box lies flat and label side up on the output shelf, the table is clear of game pieces, and both arms are home with their grippers open.

Setup

Go through both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the player row along the front edge, the board area at the front right, the bag rack, the banding cradle, the nesting spot, the rules

spot, the bank spot, the dice spot, the box base and the lid in this episode's config, the output shelf, and both home positions.

3. The inside of the box base is in view, so everything can be seen going in.
4. The left wrist camera is clean and sees the red and blue trays, the bag rack, the board and the box.
5. The right wrist camera is clean and sees the green and yellow trays, the bank tray, the dice well, the board, the box and the output shelf.
6. Both arms are at home with their grippers open.
7. Both arms reach the box base, the lid and the output shelf in this episode's config, and both arms reach the middle of the board and the banding cradle, without extending to a joint limit or folding in on themselves.
8. The left arm reaches the red tray, the blue tray, the rules spot and every bag mouth in the rack in the same way. The right arm reaches the green tray, the yellow tray, the bank tray and the dice well in the same way.
9. Nothing is in the way between the player row, the bag rack, the banding cradle, the nesting spot, the board area, the box base, the lid and the output shelf.
10. The bag rack and the banding cradle are fixed and do not slide when a gripper touches them.
11. The output shelf is fixed, holds the weight of the loaded box, and has at least 2 cm of clear space all round the place the box will stand.
12. The work surface is clean, dry, level and clear of anything not listed below.

Materials checklist

1. The board lies open and flat, playing side up, fully inside the board area at the front right.
2. The four player trays stand along the front edge in red, blue, green, yellow order, left to right. Each tray holds its four matching pawns and one face down stack of four cards.
3. The red, blue, green and yellow score markers lie side by side on the board.
4. The center tile and the first player marker are on their spots on the board.
5. The bank tray on the bank spot holds twelve coins: four one coins, four five coins and four ten coins.
6. The dice well on the dice spot holds four dice.
7. The draw spot on the board holds one squared face down stack of eight cards.
8. The rules sheet lies flat and printed side up on the rules spot.
9. Five bags stand open in the bag rack, one per slot, in the order red, blue, green, yellow, neutral, left to right. Every mouth is held open and clear.
10. One elastic band is stretched evenly around the four low pegs of the banding cradle.
11. The empty box base stands square in the packaging area, directly in front of the output shelf, and it is empty. The lid lies open side up in the lid area beside it: to the right of the box

base in **Config L**, to the left of the box base in **Config M** and **Config R**.

12. The box base is big enough to hold the whole packout with every item at least 1 cm below its rim.
13. The nesting spot, the banding cradle and the output shelf are clear.
14. Every card is flat, dry and not torn. Every bag seal and the elastic band are undamaged. The six trays nest without force.

Where things are

- **Player row:** along the front edge, left to right: the red, blue, green and yellow trays. It is empty when the trays are nested.
- **Board area:** the front right of the table, to the right of the player row. The open board lies here in every config, and it is empty when the board is packed.
- **Bank spot:** the right side of the table, behind the board area. The bank tray stands here.
- **Dice spot:** beside the bank spot, toward the middle of the table. The dice well stands here.
- **Rules spot:** the left side of the table. The rules sheet lies here until it goes in the box.
- **Bag rack:** a fixed rack with five slots at the middle of the table, behind the banding cradle. Its slots are red, blue, green, yellow and neutral, left to right. Each bag stands open in its slot until it is sealed.
- **Banding cradle:** a fixed shallow tray at the middle of the table, between the player row and the bag rack. Its four walls square the cards, and its four low pegs hold the elastic band stretched and clear of the deck.
- **Nesting spot:** a marked rectangle to the left of the banding cradle. The six trays are stacked here.
- **Output shelf:** the fixed shelf along the back edge, at the left end (**Config L**), in the middle (**Config M**) or at the right end (**Config R**). It is clear until Step 8 and holds the closed box at the end.
- **Packaging area:** the place on the table directly in front of the output shelf, wherever the shelf stands. The box base stands here from the start of the episode until Step 8.
- **Lid area:** the place beside the box base where the lid lies open side up: to the right of the box base in Config L, to the left of it in Config M and Config R. It is empty from Step 7 on.
- **Home positions:** the two clear resting poses outside every working path, grippers open.

Vocabulary

Which arm does what

- The **left gripper** owns the red and blue player pieces, the red and blue card stacks, the red and blue trays, and the rules sheet. It never picks up a green or yellow piece, a coin, a die or the first player marker.

- The **right gripper** owns the green and yellow player pieces, the green and yellow card stacks, the green and yellow trays, the bank tray, the twelve coins, the four dice, the dice well and the first player marker. It never picks up a red or blue piece.
- **Both grippers together** move the center tile, the eight card draw pile, the elastic band, the finished tray stack, the folded board, the banded deck, the lid and the closed box.
- **Sealing a bag:** one gripper holds the closed end of the seal still while the other gripper presses the seal shut along its whole length. The gripper that filled the bag is the one that presses it shut.
- The roles never swap with the config.

Words for the things

- **Player piece:** one colored pawn or one colored score marker.
- **Color bag:** the red, blue, green or yellow bag. It takes only pieces of its own color.
- **Neutral piece:** the center tile, a coin, a die or the first player marker.
- **Neutral bag:** the bag that takes all eighteen neutral pieces.
- **Card stack:** one face down group of cards before the groups are put together.
- **Deck:** all twenty-four cards squared into one face down pile.
- **Band:** the elastic loop that holds the deck. It goes round the middle of the deck, across its short way.
- **Tray stack:** the bank tray, the four player trays and the dice well nested in the fixed order.
- **Long fold:** the fold line that runs the long way across the open board. It is closed first.
- **Short fold:** the fold line that crosses the long fold. It is closed second.
- **Rim:** the top edge of the box base. Packed items stay at least 1 cm below it.
- **Seam:** the line where the lid sits over the box base.

Moves

- **Lower in:** the gripper carries the piece level over the open mouth or the open box, lowers it straight down until it is resting inside, and lets go. Nothing is dropped from above.
- **Nest:** the gripper lowers one empty tray into the tray below it until it is resting, level and steady. No tray is ever pushed or forced.
- **Square a stack:** a closed gripper tip nudges the cards against the walls of the banding cradle until every edge is in line. No card is turned face up.
- **Flat:** the item lies level, with no corner folded up and no edge riding up a wall.
- **Settled:** the thing stays put after the gripper lifts clear, with nothing rocking, sliding or tipping.

Steps

Steps 1 to 9 are one teardown. Every piece is carried level on a clear path and lowered until it is resting.

Step 1: Fill and seal the four color bags

Goal: the red, blue, green and yellow bags stand sealed in the rack, and each one holds its four pawns and its one score marker.

1.1 The red bag

- The **left gripper** closes on one red pawn in the red tray, lifts it clear, carries it level to the red bag, and lowers it in through the open mouth.
- The **left gripper** does the same with the other three red pawns, one at a time.
- The **left gripper** picks the red score marker off the board and lowers it into the red bag.
- The **left gripper** presses the air out below the seal without pushing a piece back up into it.
- The **right gripper** holds one end of the seal still. The **left gripper** presses the seal shut along its whole length.

1.2 The blue bag

- The **left gripper** moves the four blue pawns from the blue tray into the blue bag, one at a time, then the blue score marker from the board.
- The **right gripper** holds one end of the seal still while the **left gripper** presses it shut along its whole length.

1.3 The green bag

- The **right gripper** moves the four green pawns from the green tray into the green bag, one at a time, then the green score marker from the board.
- The **left gripper** holds one end of the seal still while the **right gripper** presses it shut along its whole length.

1.4 The yellow bag

- The **right gripper** moves the four yellow pawns from the yellow tray into the yellow bag, one at a time, then the yellow score marker from the board.
- The **left gripper** holds one end of the seal still while the **right gripper** presses it shut along its whole length.

Check: each of the four bags holds five pieces of its own color, no seal has a gap, and no piece is caught in a seal. If a seal has a gap, the gripper that pressed it presses it again.

Expected state: four sealed color bags stand in the rack. Each player tray holds only its four card stack. No score marker is left on the board.

Step 2: Fill and seal the neutral bag

Goal: the neutral bag stands sealed in the rack holding the center tile, twelve coins, four dice and the first player marker.

- Both grippers close on opposite sides of the center tile, lift it level off the board, carry it to the neutral bag, and lower it in flat before letting go.
- The **right gripper** moves the twelve coins from the bank tray into the neutral bag, one at a time, in the order four one coins, four five coins, four ten coins.
- The **right gripper** moves the four dice from the dice well into the neutral bag, one at a time.
- The **right gripper** picks the first player marker off the board and lowers it into the neutral bag last.
- The **right gripper** presses the air out below the seal without dragging the center tile up into it.
- The **left gripper** holds one end of the seal still while the **right gripper** presses it shut along its whole length.

Check: the bag holds eighteen pieces, the seal has no gap, and the board, the bank tray and the dice well hold no neutral pieces. If a piece is still out, the right gripper puts it in before the bag is sealed.

Expected state: five sealed bags stand in the rack. The bank tray and the dice well are empty.

Step 3: Gather the cards and band the deck

Goal: all twenty-four cards lie face down as one squared deck in the banding cradle, held by one flat band.

3.1 Gather the stacks

- The **left gripper** pinches the red four card stack at the middle of one short side, lifts it level, lays it flat in the banding cradle, and lets go.
- The **left gripper** lays the blue stack on top of the red stack in the same way.
- The **right gripper** lays the green stack, then the yellow stack, on top in the same way.
- Both grippers take opposite short sides of the eight card draw pile on the board, lift it level, and lay it on top of the four stacks.
- After each stack is laid down, one closed gripper tip squares the deck against the cradle walls.

3.2 Put the band on

- Each gripper pinches one exposed side of the elastic band on the cradle pegs.
- Both grippers lift the band evenly off the four pegs, keep it stretched, bring it over the middle of the deck, and lower it round the deck across its short way.

- Both grippers let go together. The band lies flat, untwisted, and within 1 cm of the middle of the deck.

3.3 Check the band holds

- Both grippers lift the banded deck about 2 cm, hold it level for one second, and lower it back into the cradle.

Check: the deck holds twenty-four cards, every card is face down, the edges are square, and no card slid out during the lift. If the band is twisted or off the middle, both grippers take it off and put it on again.

Expected state: one banded deck lies in the cradle. Every player tray and the draw spot are empty.

Step 4: Nest the six trays

Goal: one steady stack stands on the nesting spot, from the bottom up: bank tray, red, blue, green, yellow, dice well.

- The **right gripper** lifts the empty bank tray off the bank spot, carries it level to the nesting spot, and sets it down with its label facing the environment camera.
- The **left gripper** lifts the red tray off the player row and nests it in the bank tray. The **left gripper** nests the blue tray on top of it in the same way.
- The **right gripper** nests the green tray, then the yellow tray, in the same way.
- The **right gripper** lifts the empty dice well off the dice spot and nests it inside the yellow tray at the top.
- Both grippers close on opposite sides of the finished stack, lift it about 2 cm, hold it level for one second, and set it back down on the nesting spot.

Check: the stack is in the fixed order, every label faces the camera, no rim is caught, and the stack did not shift during the lift. If a tray sits proud, the gripper that nested it lifts it and lowers it again.

Expected state: one steady tray stack stands on the nesting spot. The player row, the bank spot and the dice spot are empty.

Step 5: Fold the board

Goal: the board is a flat four layer packet with the playing face inside.

Before folding, look over the board once: no card, marker, tile or first player marker is left on it.

5.1 Close the long fold

- The **right gripper** rests on the right half of the board near the fold line and holds it down.

- The **left gripper** pinches the middle of the free left edge, lifts it about 3 cm, and swings the left half over the fold line onto the right half.
- The **left gripper** lowers the folded half until it is resting and lets go. Then the **right gripper** lifts off.

5.2 Close the short fold

- The **right gripper** rests on the back half near the fold line and holds it down.
- The **left gripper** pinches the middle of the free front edge through both layers, lifts it about 3 cm, and swings the front half away from the front edge onto the back half.
- The **left gripper** lowers the four layer packet until it is resting and lets go. Then the **right gripper** lifts off.

Check: the playing face is inside, the four layers line up within 1 cm at every edge, and no corner stands more than 1 cm off the packet. Either gripper may press a fold line lightly or nudge an edge, at most twice in the whole step.

Expected state: the board is a flat four layer packet in the board area. The board area holds nothing else.

Step 6: Pack the box

Goal: the folded board, the rules sheet, the tray stack, the banded deck and the five sealed bags are inside the box base, in the fixed order, every one of them at least 1 cm below the rim.

Look where the box base stands before the first carry: in front of the output shelf at the left end of the back edge in Config L, in the middle of the back edge in Config M, or at the right end in Config R.

- Both grippers close on opposite edges of the folded board, lift it level, carry it to the box base, and lay it flat on the bottom with its outer face up.
- The **left gripper** pinches the rules sheet at the middle of one long edge, carries it to the box, and lays it flat, printed side up, on the folded board.
- Both grippers close on opposite sides of the tray stack, lift it level, carry it to the box, and lower it in without letting the trays come apart.
- Both grippers close on opposite short sides of the banded deck, lift it level out of the cradle, carry it to the box, and lay it flat with the card backs up.
- The **right gripper** lowers the neutral bag into the box first and lays it flat with its label up.
- The **left gripper** lowers the red bag in next, then the blue bag, each one flat with its label up.
- The **right gripper** lowers the green bag in, then the yellow bag, each one flat with its label up.

Check: the packing order was board, rules sheet, trays, deck, then the bags in the order neutral, red, blue, green, yellow. Everything is settled and at least 1 cm below the rim, and nothing hangs over it. The board area, the rules spot, the bag rack, the banding cradle, the nesting spot, the player row, the bank spot and the dice spot are all empty.

Expected state: the box base holds the whole game and stands open in the packaging area.

Step 7: Put the lid on

Goal: the lid sits square and fully down on the box base, with nothing caught in the seam.

Look where the lid lies before reaching for it: to the right of the box base in Config L, to the left of it in Config M and Config R.

- Both grippers take a balanced hold low on opposite short sides of the lid.
- Both grippers lift the lid straight up, keep it level, and carry it across to a place straight above the box base.
- Both grippers line the lid up with the box base, lower it straight down until it is resting on all four sides, and let go together. The lid is never slid across the packed things.
- If the lid stands proud on one side, either gripper may press lightly on one corner and then on the corner opposite it. No more than those two presses.

Check: the lid overlaps the base evenly, no side gap is wider than 3 mm, the lid is no more than 5 mm off square, and no bag, card, paper edge or tray rim shows in the seam.

Expected state: the box is closed in the packaging area. The lid area is empty.

Step 8: Put the closed box on the shelf

Goal: the closed box lies flat and label side up on the output shelf.

Look where the output shelf stands before lifting: the left end of the back edge in Config L, the middle in Config M, or the right end in Config R.

- Both grippers close on opposite long sides of the box, on the walls of the base below the seam, so neither gripper holds the lid.
- Both grippers lift the closed box straight up at least 2 cm, keep it level, and carry it to the output shelf without leaving the environment camera frame.
- Both grippers lower the box until its whole bottom is resting on the shelf, label side up, its long side parallel to the front of the shelf and every edge at least 2 cm inside the shelf boundary.
- Both grippers let go together and draw back without shifting the lid.

Check: the box is closed, flat, label side up, fully supported, and clear of the shelf edge. If it is crooked, both grippers lift it and set it down again.

Expected state: the closed box rests on the output shelf. The packaging area is empty and the table is clear of game pieces.

Step 9: End the episode

Goal: the game is packed and shelved, and both arms are safely home.

- Look over the table once: the closed box lies flat and label side up on the shelf, and the player row, the board area, the bank spot, the dice spot, the rules spot, the bag rack, the banding cradle, the nesting spot, the packaging area and the lid area are all empty.
- Return both arms home with their grippers open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the closed box off the shelf, take the lid off, and set the box base back in the packaging area for the next episode's config: in front of the shelf at the left end of the back edge (Config L), in the middle (Config M), or at the right end (Config R). Move the shelf to that place first. Vary the config between episodes.
2. Lay the lid open side up in the lid area beside the box base: to the right of it in Config L, to the left of it in Config M and Config R.
3. Take everything out of the box. Open the board playing side up in the board area and lay the rules sheet flat on the rules spot.
4. Unnest the trays. Stand the four player trays along the front edge in red, blue, green, yellow order, put the bank tray back on the bank spot, and put the dice well back on the dice spot.
5. Take the band off the deck and stretch it evenly round the four pegs of the banding cradle.
6. Open all five bags. Put four matching pawns in each player tray, lay the four score markers on the board, put the center tile and the first player marker on their board spots, put twelve coins in the bank tray, and put four dice in the dice well.
7. Lay the cards out again as five squared face down stacks: four cards in each player tray and eight cards on the draw spot.
8. Stand the five empty open bags in the rack in red, blue, green, yellow, neutral order.
9. Replace a card that is curled, torn or damp, a bag whose seal will not close, a band that has lost its stretch, a tray that will not nest, or a board with a torn fold line.
10. Confirm the box base is empty, the shelf is clear, every count matches the materials checklist, then run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the start timestamp, violation name, and SOP rule broken. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken. Delete and record again only for human intervention or for a failure that is not a violation.

Violations

Violation: Config misaligned

- **Visible cue:** what is on the table does not match the config being run: the box base is not directly in front of the output shelf, the lid is on the wrong side of the box base, the shelf is moved during the episode, or the box, the lid or the closed box is carried to a place other than the one for this episode's config.
- **SOP rule broken:** the start position. The shelf stands at the left end of the back edge (Config L), in the middle (Config M) or at the right end (Config R), the box base stands directly in front of it, and the lid lies beside the box base on the side the config gives. One config per episode, never changed during the episode.
- **Coaching note:** look at where the shelf and the box stand before the first carry in Step 6, and pack, close and shelve at those places.

Violation: Fixed order broken

- **Visible cue:** a step starts before the step before it has reached its end state, the bags are filled in an order other than red, blue, green, yellow, the neutral bag is filled in another order, the cards are gathered in another order, the trays are nested in another order, the short fold is closed before the long fold, or the box is packed in another order.
- **SOP rule broken:** What stays the same, and Steps 1 to 6, set the step order, the color order, the neutral order, the card order, the nesting order, the fold order and the packing order.
- **Coaching note:** name the next thing before reaching for it, and finish the color or the group you are on before starting the next.

Violation: Wrong color or wrong count in a color bag

- **Visible cue:** a color bag holds a piece of another color, is missing one of its four pawns or its score marker, holds an extra piece, or a player piece goes into the neutral bag.
- **SOP rule broken:** Step 1, each color bag holds exactly four matching pawns and the one matching score marker.

- **Coaching note:** count five pieces of one color into the bag before pressing the seal.

Violation: More than one loose piece at once

- **Visible cue:** two or more pawns, markers, coins or dice are lifted or carried in one grip.
- **SOP rule broken:** What stays the same, and Steps 1 and 2, one loose piece per pick.
- **Coaching note:** one piece, all the way in, then go back for the next.

Violation: Wrong gripper on a loose piece

- **Visible cue:** the right gripper picks up a red or blue piece, or the left gripper picks up a green or yellow piece, a coin, a die or the first player marker.
- **SOP rule broken:** Which arm does what, and Steps 1 and 2. The left gripper works red and blue, the right gripper works green, yellow and every neutral piece.
- **Coaching note:** left hand for red and blue, right hand for green, yellow and the neutral pieces.

Violation: Bag left open or sealed on a piece

- **Visible cue:** a seal has a visible gap, a piece or a fold of the bag lies across the seal, or a sealed bag opens or spills before it is packed.
- **SOP rule broken:** Steps 1 and 2, every filled bag is pressed shut along its whole length with a clear seal.
- **Coaching note:** press the air out below the seal, clear the line, then press from one end to the other.

Violation: Neutral piece missed or left behind

- **Visible cue:** the neutral bag does not hold the center tile, twelve coins, four dice and the first player marker, it holds something else, or a neutral piece is still on the board, in the bank tray or in the dice well after Step 2.
- **SOP rule broken:** Step 2, exactly eighteen named pieces go in and every source is left empty.
- **Coaching note:** clear the board, the bank tray and the dice well before the seal goes shut.

Violation: Card turned face up, bent or fanned

- **Visible cue:** a card face shows, a card is bent or fanned, or cards are moved one at a time instead of as whole stacks.
- **SOP rule broken:** Step 3, whole stacks move face down and level, and no card is turned, fanned, bent or dealt.
- **Coaching note:** carry each stack as one flat pile with a pinch at the middle of a short side.

Violation: Deck not squared or wrong card count

- **Visible cue:** the deck does not hold twenty-four cards, or card edges are left out of line in the cradle when the band goes on.
- **SOP rule broken:** Step 3, all five source stacks make one squared twenty-four card deck.
- **Coaching note:** square the deck against the cradle walls after every stack you lay down.

Violation: Band twisted, off center, or not holding

- **Visible cue:** the band is twisted, lies more than 1 cm off the middle of the deck, is put on by one gripper alone, or a card slides out during the 2 cm lift check.
- **SOP rule broken:** Step 3, both grippers lift the band off the pegs and lower it flat round the middle of the deck, and the deck passes the lift check.
- **Coaching note:** both hands, even stretch, straight down over the middle.

Violation: Trays nested wrong or stack not steady

- **Visible cue:** the order is not bank, red, blue, green, yellow, dice well, a label faces away from the camera, a rim is caught, a tray is pushed or forced, or the stack shifts during the lift check.
- **SOP rule broken:** Step 4, the fixed nesting order, labels to the camera, no force, and a steady stack.
- **Coaching note:** lower one tray at a time and stop if it does not seat on its own.

Violation: Board folded with something still on it

- **Visible cue:** folding starts while a card, a marker, the center tile or the first player marker is still on the board.
- **SOP rule broken:** Step 5, the board is clear before the first fold.
- **Coaching note:** look over the whole board once before the right gripper holds it down.

Violation: Board folded the wrong way round

- **Visible cue:** the short fold closes before the long fold, a half is folded away from the fold line, or the playing face ends up on the outside of the packet.
- **SOP rule broken:** Step 5, the long fold first, the short fold second, playing face inside.
- **Coaching note:** long fold, then short fold, and the playing face always ends up hidden.

Violation: Folded board out of line

- **Visible cue:** matching edges of the folded board are more than 1 cm out of line, or a corner stands more than 1 cm off the packet, and the next step starts anyway.
- **SOP rule broken:** Step 5, the four layers line up within 1 cm before the board is packed.
- **Coaching note:** use one of the allowed nudges to bring the edges in line before you move on.

Violation: Packing order or lie wrong

- **Visible cue:** things go into the box in another order, the tray stack comes apart, the deck goes in on its side or card faces up, a bag label faces down, or the bags go in in an order other than neutral, red, blue, green, yellow.
- **SOP rule broken:** Step 6, the fixed packing order and the way each thing lies in the box.
- **Coaching note:** build the packout from the board up, and keep every label facing up.

Violation: Something left out of the box or riding high

- **Visible cue:** a game item is still on the table when the lid comes over, or a packed item sits less than 1 cm below the rim or hangs over it.
- **SOP rule broken:** Step 6, everything goes in the box and everything sits at least 1 cm below the rim.
- **Coaching note:** look over every spot, then look across the rim, before you reach for the lid.

Violation: Lid crooked or trapping something

- **Visible cue:** a side gap is wider than 3 mm, the lid is more than 5 mm off square, the overlap is uneven, the lid is slid across the packed things, or a bag, card, paper edge or tray rim shows in the seam.
- **SOP rule broken:** Step 7, the lid is lined up above the base, lowered straight down, and left square and fully on with a clear seam.
- **Coaching note:** line up first, lower straight, and use only the two allowed corner presses.

Violation: Closed box carried by the lid or tilted

- **Visible cue:** a gripper holds only the lid, one gripper carries the box alone, the loaded box visibly tilts, or its contents shift during the carry.
- **SOP rule broken:** Step 8, both grippers hold the base walls below the seam and carry the box level.
- **Coaching note:** hold low on the base, both hands, and travel level.

Violation: Box not flat in the shelf zone

- **Visible cue:** the box rests on an edge, label side down, its long side not parallel to the front of the shelf, partly off the shelf, or closer than 2 cm to a shelf boundary.
- **SOP rule broken:** Step 8, the box lies flat, label side up, fully supported, parallel to the shelf front, and at least 2 cm inside the boundary.
- **Coaching note:** lower until the whole bottom is carried by the shelf, then check the line before you let go.

Violation: Retry limit passed

- **Visible cue:** the same piece or stack gets more than one deliberate re-grip or more than two small nudges.

- **SOP rule broken:** What stays the same sets the retry limit.
- **Coaching note:** stop fiddling. If the allowed try does not work, end the episode and reset the approach.

Violation: Dropped or knocked over

- **Visible cue:** under command, a piece, a stack, a tray, a bag, the deck, the board, the lid or the box falls, or a gripper knocks over the bag rack, the cradle, the tray stack or the box, with no gripper failure, drift or controller fault visible.
- **SOP rule broken:** Steps 1 to 8, everything is carried level on a clear path and lowered until it is resting.
- **Coaching note:** watch the whole path, not just the target.

Violation: Work outside the camera frame

- **Visible cue:** a bag mouth, the band, the tray stack, a board fold, the packing, the lid closing or the shelf placement is worked partly or wholly outside the environment camera frame.
- **SOP rule broken:** the Setup hardware checklist and the Steps, every action happens inside the camera frame.
- **Coaching note:** the work has drifted outside the framed area. Set the spots back to their setup places.

Violation: Wrong episode ending

- **Visible cue:** the episode ends before the box is settled on the shelf, with a game item still on the table, with the lid off, with an arm away from home, or with a gripper not fully open.
- **SOP rule broken:** Step 9, look over the table, then return both arms home with grippers open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Violation: Human intervention during recording (critical, delete the episode)

- **Visible cue:** a person touches or moves a game piece, a bag, a tray, the board, the box, the lid, a fixture, the shelf or the robot while recording.
- **SOP rule broken:** after setup, the robot does the whole task on its own.
- **Episode handling:** stop, delete the episode, reset, and record again.
- **Coaching note:** keep people out of the workspace until the episode has ended.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- Recording stopped or paused during the episode (recording system).
- Camera dropped frames or lost its feed (capture system).

- Hardware fault on an arm: gripper failure, drift, controller caused collision, or motor error.
- A spot sits outside an arm's comfortable reach, so the arm stalls at a joint limit or has to fold in on itself to work it. Move the spot and run the Setup checklists again.
- The bag rack, the banding cradle, the table or the shelf shifts or gives way although it was handled correctly.
- Defective object: a bag seal that splits, a band that snaps, a tray that will not nest, a card that is already curled, damp or torn, a torn or stiff board fold line, a warped lid, or a loaded box heavier than the arms are rated to carry.
- Wrong setup at episode start: a count does not match the materials checklist, a piece is missing or broken, or a spot holds the wrong things. Reset error, not a violation.

Annotation subtasks (from SOP)

1. Move one red pawn into the red bag
2. Move the red score marker into the red bag
3. Press the red bag shut
4. Fill and seal the blue bag
5. Fill and seal the green bag
6. Fill and seal the yellow bag
7. Move the center tile into the neutral bag with both grippers
8. Move one coin into the neutral bag
9. Move one die into the neutral bag
10. Move the first player marker into the neutral bag
11. Press the neutral bag shut
12. Lay one card stack in the banding cradle
13. Square the deck against the cradle walls
14. Put the band round the deck with both grippers
15. Lift the banded deck to check it holds
16. Nest one tray in the stack
17. Lift the tray stack to check it is steady
18. Close the long fold of the board
19. Close the short fold of the board
20. Lay the folded board in the box base
21. Lay the rules sheet on the folded board
22. Lower the tray stack into the box base
23. Lay the banded deck in the box base
24. Lower one sealed bag into the box base

25. Put the lid on the box base
26. Carry the closed box to the output shelf
27. Return both arms home and end the episode